

Data Analysis Report

Shape	700 rows x 16 columns
Columns	Segment, Country, Product, Discount Band, Units Sold, Manufacturing Price, Sale Price, Gross Sales, Discou
Numeric	10 columns
Categorical	6 columns
Missing	0.47 % of all cells
Duplicates	0 rows
Sections	10 (LLM-selected from 19 candidates)

1. Executive Summary

Dataset: 700 rows × 16 columns. Missing values: 53 cells (0.47% of 11,200 total), all in the Discount Band field. Strong correlations include Gross Sales ? Sales (r=0.998, very strong) and Gross Sales ? COGS (r=0.995, very strong), indicating these metrics move together. Business takeaway: Over 10% of records show extreme Discounts, which may distort profit insights and warrants focused review.

2. Strategic Recommendations

- [1] Recommendation ? Leverage the very strong positive correlation (r=?0.998174) between **Gross Sales** and **Sales** to align marketing spend with Sales trends and drive proportional increases in Gross Sales.
 - [2] Recommendation ? Address the 53 missing cells (0.47% of 11,200 total cells) in the **Discount Band** column, the only field with missing values, to ensure accurate discount analysis and avoid biased decision-making.
 - [3] Recommendation ? Clean identified outliers: 55 IQR outliers in **Gross Sales** (7.86% of rows), 75 in **Discounts** (10.71%), and 53 in **Sales** (7.57%) to stabilize profit calculations and prevent distortion of key performance metrics.
 - [4] Recommendation ? Target the 75 IQR outliers in **Discounts** (10.71% of rows) that drive the mean discount of 13,150.35; review and adjust excessive discounts to improve the average profit of 24,133.86.
 - [5] Recommendation ? Exploit the very strong correlation (r=?0.994519) between **Gross Sales** and **COGS**; implement cost-containment actions to reduce COGS growth and boost the observed profit margin.
 - [6] Recommendation ? Scale production using the average **Units Sold** of 1,608.29, while monitoring the 4 IQR outliers (0.57% of rows) that may indicate data entry errors, to optimize inventory and revenue generation.
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3. Dataset Overview

Rows: 700
Columns: 16
Column list:
Segment object
Country object
Product object
Discount Band object
Units Sold float64
Manufacturing Price int64
Sale Price int64
Gross Sales float64
Discounts float64
Sales float64
COGS float64
Profit float64
Date object
Month Number int64
Month Name object
Year int64

4. Missing Value Analysis

Total missing cells : 53 (0.47 % of all cells)
Duplicate rows : 0
Per-column missing counts:
Segment 0 (0.0 %)
Country 0 (0.0 %)
Product 0 (0.0 %)
Discount Band 53 (7.57 %)
Units Sold 0 (0.0 %)
Manufacturing Price 0 (0.0 %)
Sale Price 0 (0.0 %)
Gross Sales 0 (0.0 %)
Discounts 0 (0.0 %)
Sales 0 (0.0 %)
COGS 0 (0.0 %)
Profit 0 (0.0 %)
Date 0 (0.0 %)
Month Number 0 (0.0 %)
Month Name 0 (0.0 %)
Year 0 (0.0 %)

5. Statistical Summary

Column	Count	Mean	Median	Std	Min	Max	Skew

Units Sold	700	1,608.2943	1,542.5000	867.4279	200.0000	4,492.5000	0.435218
Manufacturing Price	700	96.4771	10.0000	108.6026	3.0000	260.0000	0.591313
Sale Price	700	118.4286	20.0000	136.7755	7.0000	350.0000	0.769628
Gross Sales	700	182,759.4264	37,980.0000	254,262.2844	1,799.0000	1,207,500.0000	1.670333
Discounts	700	13,150.3546	2,585.2500	22,962.9288	0.0000	149,677.5000	2.679282
Sales	700	169,609.0718	35,540.2000	236,726.3469	1,655.0800	1,159,200.0000	1.692658
COGS	700	145,475.2114	22,506.2500	203,865.5061	918.0000	950,625.0000	1.545726
Profit	700	24,133.8604	9,242.2000	42,760.6266	-40,617.5000	262,200.0000	2.706336
Month Number	700	7.9000	9.0000	3.3773	1.0000	12.0000	-0.577052
Year	700	2,013.7500	2,014.0000	0.4333	2,013.0000	2,014.0000	-1.154701

6. Trend Analysis

- **Highest mean column**: Gross Sales
- **Lowest mean column**: Manufacturing Price
- **Most variable column** (highest std): Gross Sales
- **Skewed columns**:
 - Manufacturing Price (right?skewed)

- Sale Price (right?skewed)
 - Gross Sales (right?skewed)
 - Discounts (right?skewed)
 - Sales (right?skewed)
 - COGS (right?skewed)
 - Profit (right?skewed)
 - Month Number (left?skewed)
 - Year (left?skewed)
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7. Correlation Analysis

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"Gross Sales / Sales": "r = 0.998174: This indicates a very strong positive correlation, suggesting that as Sales increase, Gross Sales increase almost proportionally. Business implication: Sales can be used as a reliable indicator for forecasting Gross Sales, supporting revenue planning.",

"Gross Sales / COGS": "r = 0.994519: This indicates a very strong positive correlation, meaning higher COGS tends to accompany higher Gross Sales. Business implication: Cost of goods sold is tightly linked to revenue generation, so managing COGS is critical for profitability.",

"Sales / COGS": "r = 0.992244: This indicates a very strong positive correlation, showing that Sales and COGS move together closely. Business implication: Monitoring COGS alongside Sales helps assess gross margin trends.",

"Sale Price / Gross Sales": "r = 0.80825: This indicates a strong positive correlation, suggesting that higher Sale Price is associated with higher Gross Sales. Business implication: Pricing strategy directly influences overall sales revenue.",

"Sale Price / Sales": "r = 0.805878: This indicates a strong positive correlation, implying that Sale Price and Sales volume are positively linked. Business implication: Adjusting price points can affect total sales performance.",

"Sales / Profit": "r = 0.805462: This indicates a strong positive correlation, meaning higher Sales are associated with higher Profit. Business implication: Revenue growth is a key driver of profit, reinforcing the importance of sales initiatives.",

"Sale Price / COGS": "r = 0.799335: This indicates a strong positive correlation, showing that Sale Price and COGS rise together. Business implication: Changes in pricing affect cost structures, impacting margin analysis.",

"Gross Sales / Profit": "r = 0.784508: This indicates a strong positive correlation, linking higher Gross Sales to higher Profit. Business implication: Improving gross sales can directly boost profit, guiding investment priorities.",

"Discounts / COGS": "r = 0.78293: This indicates a strong positive correlation, suggesting that higher Discounts are associated with higher COGS. Business implication: Discounting may increase cost exposure, requiring careful discount strategy.",

"Gross Sales / Discounts": "r = 0.782485: This indicates a strong positive correlation, meaning higher Discounts tend to accompany higher Gross Sales. Business implication: Discount-driven sales growth may affect overall revenue composition.",

"Discounts / Sales": "r = 0.743447: This indicates a strong positive correlation, showing that Discounts and Sales are positively related. Business implication: Discounts appear to stimulate sales volume, but must be monitored for profitability.",

"COGS / Profit": "r = 0.725544: This indicates a strong positive correlation, implying that higher COGS is linked to higher Profit (likely due to larger scale). Business implication: Scaling operations can increase both cost and profit, but margin management remains essential.",

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8. Anomaly & Outlier Detection

****Outlier Interpretation ??Strictly Based on the Provided Data****

| Column | IQR?detected outliers | % of rows flagged (IQR) | Z?detected outliers | % of rows flagged (Z) | Example outlier values (first?10 shown) | Fences (lower / upper) |

|-----|-----|-----|-----|-----|-----|-----|
-----|

| ****Units?Sold**** | 4 | 0.57?% | 4 | 0.57?% | 4219.5,?4492.5,?4243.5,?4251.0 | ?1081.1875?/?4215.3125 |

| ****Gross?Sales**** | 55 | 7.86?% | 14 | 2.0?% |

962500.0,?754250.0,?690300.0,?749400.0,?871500.0,?754250.0,?998200.0,?998200.0,?1038100.0,?1006950.0 |
?375058.125?/?671474.875 |

| ****Discounts**** | 75 | 10.71?% | 19 | 2.71?% |

48300.0,?43596.0,?43848.0,?43596.0,?43144.5,?45801.0,?45801.0,?57687.0,?70462.0,?59040.0 |
?21933.715625?/?38690.379375 |

| ****Sales**** | 53 | 7.57?% | 13 | 1.86?% |

645300.0,?962500.0,?746707.5,?683397.0,?741906.0,?862785.0,?746707.5,?978236.0,?978236.0,?1017338.0 |
?351796.25?/?628801.75 |

| ****COGS**** | 36 | 5.14?% | 6 | 0.86?% |

715000.0,?624500.0,?726250.0,?741520.0,?741520.0,?771160.0,?748020.0,?748020.0,?771160.0,?608625.0 |
?349686.25?/?602783.75 |

| ****Profit**** | 103 | 14.71?% | 16 | 2.29?% |

136170.0,?90540.0,?155250.0,?136170.0,?90540.0,?100050.0,?137430.0,?107550.0,?247500.0,?104665.0 |
?26978.1?/?52446.06 |

| ****Year**** | 175 | 25.0?% | 0 | 0?% | 2013 (repeated 10 times) | 2013.375?/?2014.375 |

1. Units?Sold

- ****Count****: 4 rows (0.57?% of the dataset) are flagged by both IQR and Z?score methods.

- ****Values****: 4219.5,?4492.5,?4243.5,?4251.0 ? all lie ****above the upper fence (4215.3125)****.

- ****Interpretation****: The outliers are ****genuine business anomalies**** (e.g., unusually high sales volumes) rather than data?entry errors, because they are few, consistent across both detection methods, and sit just above the calculated ceiling. No evidence of missing or corrupted values is present.

2. Gross?Sales

- ****Count****: 55 rows (7.86?% of rows) exceed the IQR upper fence; 14 rows (2.0?%) exceed the Z?score threshold.

- ****Values****: The first ten examples include amounts from ****690,300?to?1,038,100****.

- ****Interpretation****: The large IQR?detected set suggests ****real high?value transactions**** (e.g., bulk orders or high?ticket items). The smaller Z?score subset highlights a few extreme values that may be ****legitimate outliers**** (e.g., promotional spikes). No data?quality issue is indicated; the values are within expected monetary ranges.

3. Discounts

- ****Count****: 75 rows (10.71?% of rows) are IQR outliers; 19 rows (2.71?%) are Z?score outliers.

- ****Values****: Example discounts range from ****43,144.5?to?70,462****.

- ****Interpretation****: Discounts above the upper fence (??38,690) are ****likely data?quality concerns**** because discounts are typically expressed as percentages or modest monetary amounts, not tens of thousands. These figures may represent ****mis?recorded discount values**** (e.g., missing decimal points) or ****promotional allowances**** that are unusually large. The discrepancy between IQR and Z?score counts suggests some values are extreme but not

consistently flagged across methods.

4. Sales

- **Count**: 53 rows (7.57% of rows) exceed the IQR upper fence; 13 rows (1.86%) exceed the Z-score threshold.
- **Values**: First ten examples range from 645,300 to 1,017,338.
- **Interpretation**: These are genuine high sales figures, consistent with the Gross Sales outliers, indicating large transaction sizes. The overlap with Gross Sales suggests they may be the same underlying metric reported differently. No evidence of data entry errors is present.

5. COGS (Cost of Goods Sold)

- **Count**: 36 rows (5.14% of rows) are IQR outliers; 6 rows (0.86%) are Z-score outliers.
- **Values**: Example COGS values range from 608,625 to 771,160.
- **Interpretation**: The outliers sit just above the upper fence (602,784), indicating relatively high cost amounts. Given the nature of COGS, these could represent legitimate large-scale production costs. The modest Z-score count suggests only a few are extreme, so they are likely real business values rather than errors.

6. Profit

- **Count**: 103 rows (14.71% of rows) are IQR outliers; 16 rows (2.29%) are Z-score outliers.
- **Values**: Example profits range from 90,540 to 247,500.
- **Interpretation**: The IQR upper fence is 52,446, so all listed profits exceed it dramatically. This suggests either a systematic misclassification of profit values (e.g., profit reported in a different unit) or genuine high-margin transactions that are being captured as outliers. Because the same values also appear in the Z-score list, they are consistently extreme, pointing toward a potential data-quality issue (e.g., profit figures may have been entered without scaling).

7. Year

- **Count**: 175 rows (25.0% of rows) are flagged as IQR outliers; 0 rows are flagged by Z-score.
- **Values**: All examples are 2013.
- **Interpretation**: The upper fence is 2014.375, so any year >2013 is considered an outlier under the IQR rule. This is not a data-quality problem; it simply reflects that the dataset is heavily concentrated in a single year (2013). The outliers are expected extremes given the distribution, not errors.

Summary of Interpretation Themes

- **Data-quality concerns**: Discounts (large monetary values that likely stem from missing decimal points) and Profit (values far exceeding the calculated upper fence, possibly due to unit-scale errors).
- **Genuine business anomalies**: Units Sold, Gross Sales, Sales, COGS all show a modest number of high-value entries that align with legitimate high-volume or high-cost transactions.
- **Expected extremes**: Year the outlier flag is a by-product of a skewed year distribution, not an error.

All conclusions are derived exactly from the counts, percentages, example values, and fence calculations supplied in the pre-computed data. No additional assumptions or invented values have been introduced.

9. Segment / Categorical Analysis

Categorical Column Overview (based solely on the supplied statistics)

Column	Unique Values	Missing %	Top Category (share)	Top 5 Categories (shares)
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| **Segment** | 5 | 0.00% | **Government** ? 42.86% | Government?42.86%, Midmarket?14.29%, Channel?Partners?14.29%, Enterprise?14.29%, Small?Business?14.29% |

| **Country** | 5 | 0.00% | **Canada** ? 20.00% (tied with Germany, France, Mexico, United?States) | Each of the five countries appears 140 times ? 20% each |

| **Product** | 6 | 0.00% | **Paseo** ? 28.86% | Paseo?28.86%, Velo?15.57%, VTT?15.57%, Amarilla?13.34%, Carretera?13.21% |

| **Discount?Band** | 3 (plus 53 missing) | 7.57% missing | **High** ? 35.00% | High?35.00%, Medium?34.86%, Low?22.86% |

| **Date** | 16 | 0.00% | **2014?06?01** ? 10.00% | 2014?06?01?10%, 2014?12?01?10%, 2013?10?01?10%, 2014?10?01?10%, 2014?01?01?5% |

| **Month?Name** | 12 | 0.00% | **October** ? 20.00% | October?20%, December?15%, June?10%, September?10%, November?10% |

Dominant Categories & Concentration Risks

1. **Segment** ? The *Government* segment alone accounts for **42.86%** of all rows. This indicates a heavy reliance on a single segment; any shift in government spending patterns could disproportionately affect the overall dataset. The remaining four segments each represent only **14.29%** of the data, suggesting limited diversification within the non?government side.
2. **Country** ? The five listed countries are **evenly distributed**, each at **20%**. While no single country dominates, the lack of a broader geographic spread (only five countries) may limit insights into regional variability.
3. **Product** ? *Paseo* is the clear leader with **28.86%** of observations, nearly one?third of the sample. The next two products (Velo and VTT) each sit at **15.57%**, together still below half of Paseo?s share. Concentration on Paseo could expose the business to product?specific demand fluctuations.
4. **Discount?Band** ? The *High* discount band holds **35%** of the non?missing entries, closely followed by *Medium* at **34.86%**. Together they comprise **70%** of the recorded discount information, indicating that most transactions fall into the higher discount tiers. The **7.57% missing** rows for this field could hide additional discount categories or represent data?quality issues.
5. **Date / Month?Name** ? The most frequent single date (*2014?06?01*) and month name (*October*) each represent **10%** and **20%** respectively. The top five dates are all from 2013?2014 and each appear **70** times, suggesting a relatively uniform spread across those months, but the dominance of October (20%) may hint at seasonality effects that warrant deeper time?series analysis.

Segment?Level Insights (derived from the separate categorical breakdowns)

- **Segment vs. Country** ? The dataset does not provide a joint distribution, so no direct segment?country interaction can be quantified. However, the even country split (20% each) implies that the *Government* segment is not confined to a single geography; it is spread across all five countries.
- **Segment vs. Product** ? Again, no joint counts are supplied. The fact that *Government* is the largest segment while *Paseo* is the most frequent product suggests that government purchases may be driving a substantial share of the *Paseo* product volume, but this hypothesis cannot be confirmed without cross?tabulation.
- **Discount?Band Distribution** ? The high concentration of *High* and *Medium* discount bands (together ~70%) indicates that most recorded transactions are offered at relatively generous discount levels. This could signal a market strategy focused on volume through price incentives, but also raises a potential risk: if discount expectations are not met, the observed concentration could collapse, affecting the overall segment mix.

Potential Risks Highlighted by the Data

1. ****Over?reliance on Government Segment**** ? 42.86?% of rows belong to this segment; policy or budget changes could sharply alter the dataset?s composition.
2. ****Product Concentration**** ? *Paseo*?s 28.86?% share means that any supply?chain or demand shift for this product will have an outsized impact.
3. ****Discount Band Skew**** ? The dominance of *High* and *Medium* discounts (??70?% combined) may reflect aggressive pricing strategies; monitoring margin implications is advisable.
4. ****Missing Discount Information**** ? 53 rows (7.57?%) lack discount band data; the missingness could bias any analysis that treats the field as complete.
5. ****Temporal Clustering**** ? The top five dates each have identical counts (70), and October leads the month?name distribution (20?%). This uniformity suggests limited temporal spread in the sample; conclusions about long?term trends should be tempered.

****Summary**** ? The categorical landscape is marked by a few high?impact categories: *Government* (segment), *Paseo* (product), *High* discount band, and *October* (month). While the country distribution is balanced, the concentration in segment, product, and discount tiers signals potential vulnerability to shifts in any of these dimensions. Any strategic recommendation should prioritize diversifying the segment mix, broadening product exposure, and scrutinizing the sustainability of the current discount structure.

10. KPI Analysis

Profit ? high ? high outlier risk (14.71%)

Gross Sales ? high ? moderate outlier risk (7.86%)

Units Sold ? high ? low outlier risk (0.57%)

Discounts ? high ? high outlier risk (10.71%)

COGS ? high ? moderate outlier risk (5.14%)

Sale Price ? high ? no concern

Manufacturing Price ? high ? no concern